Project Progression Record
Level 3 Extended Project Qualification

This form should be completed by the learner. Expand the boxes as needed

Name: Henry Field	Centre: HRSFC
Candidate no: 1431	Centre no: 22147

Current programme of study:

Qualification Type GCE	A Level	Subject • Mathematics • Further Mathematics • Computer Science • Physics

Date Project started: 30/09/24
Project Topic: NeuroNote - Exploring the Intersection of Cognitive Science and Computing
Rationale: *why you want to do this project and how it links to progression plans/current courses/personal interest.* From starting the EPQ program I knew I wanted to combine both an innate passion of mine and a topic I had some experience of before with a completely new subject area that had intrigued me for a long time but hadn't naturally fallen into my lap throughout my time in education so far. This project integrates both ideas seamlessly and enables me to achieve a third goal I never even knew was possible: to produce a tool that I can use throughout my life to help me organise my mind, achieve greater academic success and help others in the process with a platform I designed from the ground up. For the first of the two topics: computer science has always been a huge passion of mine for its ability to turn seemingly meaningless words and phrases strung together in somewhat arbitrary stories into a beautiful tool that doesn't reveal itself until compiled and run. From my first experience of rudimentary coding in primary school I have been hooked and have made small software tools and little coded helpers ever since. I knew the EPQ program would be an opportunity to produce a far more complicated program over a much longer period of time than I'd had the freedom of before and that I would learn so many skills along the way. My future goals and aspirations focus around a computer science centred world and with my university aspirations focused on the subject I additionally hope to use my EPQ to showcase my new skills. Simultaneously, I have been deeply fascinated by the workings of the human brain - how we process vast amounts of information and stores a seemingly infinite reservoir of memories. While I know the brain is incredibly powerful and sophisticated at understanding the world around us, I also acknowledge it does have its limitations. As a student, I often struggle to structure my thoughts and academic learnings so that I can tap into that knowledge during examination periods. With folders full of notes, online textbooks a dozen and a seemingly ever-growing tower of work I

should reflect over (as I know this to be a crucial revision resource) I knew I needed to find a way to organise my knowledge in a way that would both satisfy my need to achieve high exam grades and that appeals to my minds natural tendency to "over-organise" materials which often results in procrastination and wasted time. This has made me even more curious about cognitive function (especially of students) and how we can augment our minds.

Merging these two passions felt like a natural step, leading me to create a tool that enhances my studying while exploring the captivating world of human cognition. I aim to build the tool I (and many of my fellow peers) wish we had throughout our educations with the hope to utilise this tool for the rest of my time in sixth form and through university and beyond. This project not only helps me augment my own learning process but also deepens my understanding of both computer science and the brain's mechanisms, reinforcing my passion for both fields to create a unique project I can be truly proud of.

Aims and Objectives:

What outcome(s) do you hope to achieve?

What skills are you aiming to develop?

1. Research skills regarding understanding and learning from complex academic papers
2. Learning skills that I can use throughout other aspects of my education and further life
3. Recording best practices while documenting the outcomes of my project.
4. Software development skills: working towards a fixed end goal whilst integrating my research
5. Evaluation skills: using reflection on my work to further drive my progress and improve my future outcomes

How will you develop these skills?

A key aspect of my project is researching effective note-taking methods. As the project progresses, I will develop the very skills I am studying, creating a self-sustaining learning process. I will develop these skills by:

- Researching Effective Methods: Continuously exploring and analysing various note-taking techniques to refine my approach.
- Engaging with Experts and potential users: Reaching out to those in relevant fields to learn industry-standard practices and gain their insights.
- Iterative Improvement: Regularly reviewing and reflecting on my past prototypes to identify areas for improvement, ensuring each iteration is more effective than the last.

Initial Planning

What range of resources might you use/what types of research might you complete……e.g. specialist materials/workshop space. Books/academic journals etc. How will you ensure a good range?

My project naturally divides into three key sections: human learning, software development, and User Experience (UX) Design. Within my project all are connected by a central idea - what makes software not only functional but also enjoyable to use.

For the software development aspect, I require a suitable development environment. Fortunately, I already have the necessary IT hardware and software, acquired through

previous projects. Given the rapid pace of advancements in this field, online resources such as YouTube Tutorials, Documentation and Podcasts will likely prove more relevant than traditional media, though I will not disregard the latter.

The human learning aspect of my project requires no technical hardware but will rely heavily on research conducted over an extended period. This field focuses on knowledge that predate even the earliest methods of recording information, offering a wealth of foundational knowledge to explore in more traditional mediums such as academic journals, peer reviewed scientific research and primary research conducted with the participation of my fellow peers and those in the industry of academia.

Lastly, to learn more about user experience I will utilise primary research conducted on pre-existing software platforms to learn both from what they do well, and which areas could be improved. I will then supplement this with continual conversations, feedback sessions and surveys with my 'primary users' - those (like myself) who are eager for the platform to be a success.

I will ensure that throughout my project I utilise a wide array of sources to validate each decision I make. This use of multiple viewpoints and opinions will enable me to produce a tool that is applicable to a wide an audience as possible.

What factors might you need to take into account when planning how to manage your project?

When planning how to manage my project, several key factors need to be considered to ensure its success. First, I must clearly define the scope and objectives of the project, outlining specific goals and deliverables to stay focused and avoid scope creep.

Establishing a realistic timeline with milestones and buffer time (time not dedicated to a specific task and can be used to absorb delays / overflows in other areas of my project) is essential to keep the project on track. I also need to assess resource availability, including the necessary tools, hardware, software, and informational materials, while considering budget constraints and prioritizing cost-effective options.

Skill development will play a foundational role, as I will need to allocate time to learn new techniques, technologies and research methods as well as implement them in a variety of prototypes to best understand their applications.

I will also need to be mindful of my other A level (and general life) commitments as I unfortunately cannot dedicate 100% of my time to the EPQ program and need to plan accordingly. One initiative I am taking now however is to 'hit the ground running' during my early time in Y12 as I do not yet have many deadlines or other obstacles to work around. I aim to dive into my first few prototypes very shortly once I have completed the relevant research.

Whilst I am keen to get ahead now, I will ensure that I plan for a drop-off in time allocation over the coming months and maintain reasonable and achievable targets over the entire project.

How will you plan to bring your project in for the deadline? You should create a timeline.

To ensure my project is completed by the deadline, I will develop a detailed diary and timeline with clearly defined milestones for each phase of the work. I will prioritize tasks based on their importance and complexity, tackling the most critical components first to prevent myself having to rework aspects later (when it would be far more complex to do so). To do this I will create an Eisenhower matrix of goals and their priorities. Additionally, I have already created my initial timeline which outlines a number of goals and throughout the course of this project I will re-evaluate to ensure I hit internal school deadlines and the final submission in September.

Regular progress reviews will help me stay aligned with my goals and identify areas requiring adjustment and, by breaking the project into manageable sections and setting short-term goals I can ensure I meet my targets. Additionally, I will streamline my workflow by leveraging efficient tools, minimizing distractions, and allocating time specifically for skill-building and research. If challenges arise, I will adapt by reassigning resources and adjusting less critical aspects of the project, ensuring that my core objectives are achieved on time.

Supervisor
Signed Date

Mid Project Review

Date: 28/3/25

How much progress have you made in meeting your outcomes?

Currently in the EPQ process I feel very confident and proud of my work achieved. I am quickly approaching a functional prototype of my artefact which I can share to a wide range of people who have approached me asking to try out my software. My diary and various documents are almost entirely up to date, and I have been regularly (and thoroughly) documenting my learnings, progress and challenges throughout. I am currently ahead of my timeline by around 3 weeks and have been able to include a number of 'stretch' goals in my prototyping. I have also explored a wide range of subject adjacent material including a fascinating article written by a teacher about her experiences of various note taking methods among her students.

Overall, I find myself in a very strong position to reach my desired outcome and am excited to complete a minimum viable prototype (MVP) in the coming weeks.

How has your project changed or evolved?

At the very start of my project, I was fortunate to have a two-week work experience with an incredible Cambridge startup. From this I knew I wanted to complete a computer science adjacent topic which mixed my love of computer science with research into human nature and how we study. Initially I approached this from a written dissertation route however this has now evolved to be a prototype software project putting my research and skill development into practice. I chose to shift towards an artefact style project as I felt this would help me to stand out among my peers within the computer science industry as I am not just completing the research I would have approached if making a dissertation but also actively applying that knowledge into a functional tool which I can share with others to also help them (and in return give me feedback to further improve).

Throughout my time as a student in a variety of different education establishments, I have used a wide range of digital notetaking tools and am still left unsatisfied with the options available to me. Whilst some tools such as OneNote are very simple and for the most part very reliable, they lack tools such as rich tagging and workspaces. Whilst more complex tools such as obsidian have these features, they trade them with simplicity making their interface clunky to use and reliant on community extensions for what I consider to be core functionality.

With this knowledge in mind, I decided to build the note taking app I wanted to use throughout my time in education. Including rich filtering and sorting mechanics but made intuitive to use and approachable for a typical student (or anybody in life).

What have been the strengths/weaknesses with your project management so far?

My greatest strength so far would be my use of task prioritisation over the course of my prototype development and research. This has enabled me to maximise my time spent. An example of this is a chemical molecule editor/viewer a friend asked if I could implement. I spent about 15 minutes completing some research on various tools to help me achieve this but decided to 'shelf' the idea for the time being as the overall impact would be limited (at most 5 users) but would take me around 20+ hours to develop. Whilst I would love to have developed this (and so many other tools) I know that I first need to achieve the goals I set out during my project choice / specification stage and that, only once that is completed can I look to achieve these additional targets. This use of task prioritisation has enabled me to stay fully on course with my timeline so far, as new ideas become evaluated and slotted into open time slots in accordance with their impact level.

I would however say my biggest challenge has been getting the balance between working on code and writing up my diary / completing research pieces. Whilst I enjoy both aspects, I have occasionally found myself allowing my diary to fall behind as I ran forward with code. To prevent this, I have started allocating time goals for each coding task, stopping if I overrun which both prevents wasted time and allows me to step away, consider the problem at hand, and tackle it later with a fresh mindset. I have found that this, in addition to resulting in a more balanced project, also makes my development more efficient and hence produces a stronger end output

How are you evidencing the research you have done and how you are managing your project?

To manage my project effectively I have utilised a range of time management tools such as Timelines and Eisenhower matrixes which enable me to allocate my time to tasks that need completing most urgently to maximise the end impact of my project. Project management analysis can be found across the Planning chapter and my Diary chapter which also includes in depth analysis of my project throughout the year so far.

At the start of my project, I quickly dove into more generalist research in two discrete sections: the tools and technologies I would need to make an enjoyable user interface and the best approaches we can use for notetaking. The former yielded a wide range of resources which I explored throughout my project (and will continue to explore until conclusion). The latter was simpler to plan through, and I spent time annotating a range of peer reviewed research pieces I found through research databases, watching videos produced by influencers and my own primary research. My evaluation and notes for all these sources can be found in my Research chapter below.

The prototypes I have produced so far additionally showcase the skills I have developed from my research and my jump in progress has been very rewarding to reflect on. These can be found in the Prototypes chapter of my documentation below.

What do you still need to do?

Whilst I am now very close to my target MVP there are still many improvements and feature additions I want to make before I consider my artefact 'complete'. These include some security enhancements before going live, some 'bugs' to fix and some branding to complete.

I additionally have a few areas to further develop within my documentation such as my source analysis table: whilst all my sources are logged with references, I need to finish evaluating them for their accuracy and authority.

In the near future I also need to produce a 7-minute video which I am taking as an opportunity to 'pitch' my artefact to family and peers in an approachable infographic format. Additionally, I will have a range of documents to produce before submitting my project such as completing this PPR, completing my final evaluation and overall tying up all of the incredible work I have done to demonstrate the progress I have achieved.

I am also planning to produce a longer video showcasing my project from a first-time user's point of view in order to demonstrate many of the advantages NeuroNote has compared to its competitors in the industry.

Once these tasks are all complete, I will re-read over my entire project checking for accuracy and improving my presentation throughout to maximise clarity and information retrieval.

Supervisor
Signed: Date:

Response to advice from Mentor (at Progress Review points and in 1-1 conversations)

Log your response to verbal advice or assessment from your mentor: what will you change as a results

Throughout the EPQ Process I have had lots of very useful advice from my mentor:

The first occasion was during my transition phase from a Dissertation to an Artefact style project. I was keen to utilise my findings from a very analogue, traditional medium (the psychological study of learning) in a contrastingly different, high-tech implementation as a consumer facing web application. Within my studies I am currently completing a Computer Science A Level. After discussion with my mentor (and CS subject teachers) we agreed that my proposed EPQ project would be significantly outside the scope of the course, utilising vastly different technologies, concepts and applications at a far higher level than the A level course. To further ensure my project is distinct from the course I will be utilising research on a variety of topics such as user experience, note-taking and the structure of our minds as a focal point.

In progress meetings we discussed my paperwork completion and reflections. I am very happy with progress so far however I need to ensure I complete my source analysis table as I currently have a number of gaps. Once that is completed, I am making very strong progress in my EPQ. Additionally, I need to make sure I update my PPR and evaluations as I continue to demonstrate my progress.

In lessons we spoke about how to best showcase my skill development. Currently this is all logged within my diary entries however my mentor suggested I extract these into a more condensed skill development section which I have now completed. This section collects all of my Skill Audits throughout the project and also has a summary of my skill growth for each of the 20 skills I have tracked.

Shortly before the summer break, I met with my mentor again to discuss my project progress. She was very impressed with the state of my documentation and presentation. We spoke about the optimal structure for my project and how I can best showcase my findings. She also suggested I

produced a “walk through” video to hand in as my project artefact as my software/website alone cannot be submitted. Once I have completed my software development, I will therefore record myself exploring the interface whilst providing a commentary of decisions I made and how these benefit the end user.

End of Project

Evaluation of marketplace presentation in light of peer feedback	With my Presentation now complete and feedback gathered I am incredibly proud of the video presentation I put together. It seems that my viewers agree that I was confident in my delivery and that the video was both aesthetically pleasing and interesting to watch which makes the time and effort I put into post-production (editing and rendering) very rewarding. I seem to have struck the balance between presentation speed and content covered within the 7-minute time limit well and given a thorough insight into the story behind my project. Although there was no more time to insert other sections I do however, wish I could have covered some of the technologies I researched and implemented in more depth as they form a large part of my time during the EPQ process and my AO2/3 targeted work. Additionally, I would have liked to cover a typical users experience of using my platform however I am going to produce a second “walkthrough demo” video to submit which fills this gap. From reading through my feedback, some also mentioned commenting on the future aspirations for my project in more detail and that is both very pleasing to hear (implying they believe I may have a larger future ahead of me) and has made me think about where I can take NeuroNote. Within my planning chapter below I have now added a section for my future plans including further goals I can achieve and how I want to interact with the software development community to potentially collaboratively work on my project. Many viewers commented on the smooth visual transitions and music helping them to easily digest the information being presented, and the captions were also highly appreciated by some (with many commenting on their accuracy being “surprisingly good” - this is due to me manually writing them [from my script] and them not being automatic transcriptions as is more commonly used). Some feedback forms mentioned “including more content on the screen when talking”. Whilst I can understand the benefit of this, I do think it would have negatively harmed the visual appeal of my video. From my presentation research of both software development companies (such as Notion or Evernote) and other successful EPQ presentations it is clear to me that a smooth, well-rehearsed voiceover can provide so much more valuable content when aided by carefully chosen imagery and videos than a screen filled with text and a presenter struggling to justify it all. For further evaluation of my presentation please visit the Presentation chapter of my documentation below. Additionally, you will find insight into the presentation production journey and the feedback I gathered from a survey sent to viewers.

Evaluation of the whole project	With my final prototype now released to the public and feedback gathered through forms, meetings with users, and online comments. I have been overwhelmed by the positive reception of my work. My initial aim was to build a tool that I personally would find useful for organising my notes, while also benefiting others with similar needs. Based on the feedback and reflections throughout the project, I feel that I successfully achieved this goal. To evaluate my project effectively, I have considered each assessment objective in turn. AO1: When I started my project, I didn't yet fully appreciate the amount of work required to successfully complete an EPQ to this level, this could have caused me to miss key deadlines and fall behind however, foreseeing this could be the case, I did ensure that my initial timelines were very high level on day one. This enabled me to spend a couple of weeks researching the various sections within my project such as technical development, User interface design and most crucially: techniques to support human learning. Only once I knew enough to produce a more in-depth task sheet of research pieces to produce and prototypes to create did I develop more detailed timelines. Additionally, I utilised my time at the start of the year well, maximising my early productivity to give myself more flexibility to move targets later down the line as school assessments and other life tasks arose. My use of dedicated EPQ time within my calendar was perhaps one of my best decisions in reflection as it gave me a time and space to focus fully on my project development for the week, rather than slotting it between my other subjects where I may not necessarily be in the correct mindset to be most productive. As I mentioned earlier, I would say my greatest weakness in the planning process would be not separating research and prototyping enough within my time allocation. I naturally found myself pushing to complete prototype pieces to a high standard when that time could have been better spent conducting research on other areas and producing much simpler, less polished examples. Whilst I did manage to complete each prototype I planned for, on occasions I became a little more time constrained than I would have liked. Throughout my project however I did improve on this by utilising a range of timekeeping methods. One technique I found particularly helpful was to allocate myself a set amount of time for each task (such as 12 minutes to add a shortcuts feature) and If I were to reach the time limit without completion, I would document it and only pick up the task again after completing a research / other piece of documentation. This allowed me to ensure I kept a good balance of time allocation but also had another benefit in allowing me to re-approach the challenge with a fresh viewpoint resulting in me often solving the issue in a much shorter period of time then if I had continued to troubleshoot. This is one of the many lessons that the EPQ program has taught me which will greatly benefit me in life. AO2: I am very pleased with the wide array of sources and resources I made use of during my project. Entering the project, I knew that a wide variety of viewpoints would help me to establish best practices and form my opinions from the

	consensus of many industry experts. I didn't however appreciate how large a role primary research would play, initially I had seen it as a supplementary process to my main academic research however I have now learnt that when designing a project to be used by those around you, their opinions should heavily influence the decisions you make. It is the skill of the engineer / researcher to utilise their research skills to help establish systems and designs that can accommodate the needs of the specification they are provided. From my initial project specification produced during the planning phase, I split my research goals out into 3 primary sections: Technical documentation for my software development, Primary research to explore user experience design and, more traditional, academic research diving into reports and scientific journal to learn more about skills we can utilise to improve our studying ability. This separation of concerns benefited me nicely as I was able to efficiently split my research time depending on the resources I had at the time. For example, I found that on my train commute to school and back I was able to study a range of articles and reports when a lack of cellular connection would have previously rendered that time suboptimal. Equally, whilst out running I was able to listen to podcast hosts explore their thoughts on the subject matter when I wasn't able to read whilst on the move. I also found myself utilising a variety of techniques to record my thoughts on sources throughout the process such as: short bullet point ideas, notating areas of studies I found interesting, longer form analysis and prototyping of technical ideas. This research also allowed me to integrate proven learning techniques into the software, strengthening the theoretical foundation of my artefact. A03: Throughout the EPQ process I have found myself developing a wide array of skills as I researched brand new topics, developed software with technologies completely new to me and, produced a final report larger than any document I've worked on before. It has truly been an amazing experience and invaluable for my personal progression. To help track my skill development I recorded a number of audits at crucial milestones which tracked 20 skills I identified as being key to my success. As my research grew and my practical application of these skills developed it was very rewarding to see my skill charts represent a substantial growth over the course of the project. One of the most positive techniques I used to improve my skillset was practice via repeated implementation of the skill (such as evaluation and market research) and prototyping (in the case of technical skills) as these opportunities to put my learnings into practice enabled me to better grow my understanding. As I identified throughout my diary, as I reached issues in the development of my artefact I was able to both improve my ability to troubleshoot and research a challenge, but also, acquire new skills I hadn't even thought of before. An example of this is the difficulties I had enabling my home-based server to be accessed from the wider internet. Although it took me a number of hours to resolve, I learnt all about the structure of the world wide web and how to use tools such as Cloudflare Tunnels and Tail Scale to achieve my goals. Without having prototyped these areas I would never have come across these incredible

	tools and would have made personal projects I am now moving to work on much more challenging. AO4: My final artefact, NeuroNote has had an amazing reception by users, who appreciated its integration of complex techniques within a simple, intuitive interface. My Feedback highlights accessibility, usability, and the seamless integration of typed and handwritten notes as being “fan favourite” features. Users also valued features such as excellent typed maths support, automatic code syntax highlighting and may more note structuring features such as collapsible dropdowns. Despite these successes I appreciate the project has limitations. Dense graphs can sometimes become complex to navigate if notes do not share enough distinct categories, and optimising performance across different devices remains a challenge - especially on portable devices such as phones. Future improvements could include enhanced filtering options, offline note support and additional user onboarding features to simplify the initial experience. This evaluation process itself, gathering feedback, analysing results, and refining the artefact has taught me valuable lessons about iterative development and the importance of user-centred design. It reinforced that the effectiveness of a software tool is not solely determined by its technical sophistication but by how well it meets a user’s needs in practical contexts. Overall, my project demonstrates that combining psychological research with thoughtful software design can meaningfully improve study techniques. It also highlights my growth in planning, research, development, and evaluation skills, all of which will benefit me in future academic and professional endeavours. In conclusion, NeuroNote successfully addressed my initial research question: How can we improve our ability to study information? Through integrating academic theories, research, and iterative development, my artefact provides a practical solution to common note-taking and organisational challenges. Beyond the artefact itself, the project allowed me to develop a robust set of technical and transferable skills, deepened my understanding of research methods, and confirmed my ambition to pursue a career in computer science. The process of reflection, feedback integration, and continuous improvement has been invaluable, making this EPQ an incredible learning experience. For further evaluation of my project and the feedback I received from my peers, please visit the Evaluation chapter of my documentation below in addition to my Artefact commentary in the next chapter.
Outcome	**Evidence** – where can it be found?
Exploration of a range of ideas for your project	My Mind maps and initial project ideas can be found in the Project choice Chapter of my document below.
How you have managed your time	This can be found throughout my Diary and within my Planning Chapter.

Use of a wide range of research and resources	This can be found throughout my Research Chapter below and additionally from my bibliography at the end of this document.
Evaluated research and selected appropriate research	Sources used and my commentary / notes from them can be found in my Research chapter (within my research pieces and overall source analysis table). Their applications and further analysis can also be found in my Prototypes and Evaluation chapters.
How you made decisions and solved problems	This is found throughout my project but best concentrated within my weekly diary entries (within the Diary Chapter).
How you used technology – where appropriate	With an artefact focused on the applications of digital technology this is ingrained throughout my project. Some examples of this include: the various software suits I have developed within my Prototypes and final artefact development sections such as VS Code, my use of open-source platforms such as Zotero for my research and my heavy use of specialised video editing / 3D rendering software such as Blender and Premiere Pro to produce my custom visual effects and animated presentation video.
Presentation given to an audience	My full presentation video can be found in the attached supporting evidence area. The comments and feedback I received from peers and my production journey I undertook can all be found within my Presentation chapter below.
Making links to HE/Career	With aspirations to work within Computer science throughout my life I Have referenced these links in many sections such as my Project Choice, Diary, Skill Development and Evaluation chapters.
How you have reflected on your project management skills and the quality of your outcome.	I have reflected on my project overall throughout this PPR with a specialised focus on my end artefact being found within the Evaluation chapter below and the variety of feedback and conversation analysis I completed within.
On-going evaluation	Throughout my project I continually evaluated my progress. This is best found within my Diary and Planning chapters where my ongoing progress would help dictate my next steps.
Final Evaluation of project and presentation	This can be found within this PPR and my Evaluation and Presentation chapters.
Non dissertations– 1500 word supporting statement	Please find my 1500-word Project commentary within the chapter immediately following this document.

Date of Submission: 03/09/2025
Student Signature:

Supervisor Signature: